

Cryptography & Information Security

Study Guide

Part 1: Past Papers

MIDTERM 1 / SESSIONAL-I FALL 2025 • QUESTION 1

Q: What is the biggest disadvantage of symmetric encryption?

- a) It requires more complex and time-consuming calculations.
- b) The secret key must be transmitted securely between parties.
- c) It uses a less secure encryption function.
- d) It is no longer in use.

A: b) The secret key must be transmitted securely between parties.

MIDTERM 1 / SESSIONAL-I FALL 2025 • QUESTION 2

Q: Which statement best describes the role of a key in encryption?

- a) The key determines which encryption function is used, making decryption difficult if the function is unknown.
- b) The key contains a secret encryption function along with its parameters. A password is required to activate the key.
- c) The key only contains secret parameters used for both encryption and decryption.
- d) The key allows users to avoid reinstalling encryption software when technology changes.

A: a) The key determines which encryption function is used, making decryption difficult if the function is unknown.

MIDTERM 1 / SESSIONAL-I FALL 2025 • QUESTION 3

Q: Referring to the block cipher modes of operation, mention one benefit and one drawback of the Cipher Block Chaining (CBC) mode.

A:

- **Benefit:** Identical plaintext blocks produce different ciphertext blocks because each plaintext block is XORed with the previous ciphertext block (or IV) before encryption, hiding structural patterns.
- **Drawback:** Encryption cannot be parallelized because the encryption of each block depends on the ciphertext output of the preceding block.

MIDTERM 1 / SESSIONAL-I FALL 2025 • QUESTION 4

Q: Recall that DES Expansion P-box operates on a 32-bit input. Suppose we are working with a smaller version of DES in which expansion is performed on a 16-bit input. Expand the input block 0010 1100 0011 0100.

A: Using the provided 16-to-24 bit expansion table mapping, specific bit positions from the 16-bit input are duplicated according to the expansion permutation matrix to produce the expanded 24-bit output block.

MIDTERM 1 / SESSIONAL-I FALL 2025 • QUESTION 5

Q: Given the state matrix (in hex) for an AES encryption round:

1B	22	34	56
4A	FF	33	E3
2B	CD	4C	5B
22	AA	CC	BB

Perform the SubBytes operation followed by ShiftRows. Show the intermediate output after SubBytes and the final matrix after ShiftRows.

A:

Intermediate Output (after SubBytes lookup):

AF	9C	A4	71
D6	B3	29	D8
2B	FE	D7	AB
76	72	C0	B0

Final Output (after ShiftRows):

AF	9C	A4	71
B3	29	D8	D6
D7	AB	2B	FE
B0	76	72	C0

MIDTERM 1 / SESSIONAL-I FALL 2024 • QUESTION 6

Q: In the DES algorithm, the round key is _____ bits and the Round Input is _____ bits.

- a. 48, 32
- b. 64, 32
- c. 56, 24
- d. 32, 32

A: a. 48, 32

MIDTERM 1 / SESSIONAL-I FALL 2024 • QUESTION 7

Q: Draw the block diagram of a DES single round and properly label all components.

A: The diagram splits the 64-bit block into L_{i-1} (32 bits) and R_{i-1} (32 bits). R_{i-1} passes through Expansion Permutation (32 to 48 bits), XORs with the 48-bit Round Key K_i , enters Substitution S-Boxes (48 to 32 bits), passes through Permutation P (32 bits), and XORs with L_{i-1} to produce R_i . L_i receives R_{i-1} directly.

MIDTERM 1 FALL 2023 • QUESTION 8

Q: Which block cipher mode does not allow parallel encryption of blocks?

- a. Cipher Block Chaining (CBC)
- b. Counter mode
- c. Electronic Code Book (ECB)
- d. All of these

A: a. Cipher Block Chaining (CBC)

MIDTERM 1 FALL 2023 • QUESTION 9

Q: In the DES algorithm the Round Input is 32 bits, which is expanded to 48 bits via _____.

- a. Scaling of the existing bits
- b. Substitution of the existing bits
- c. Addition of zeros
- d. Addition of ones

A: a. Scaling of the existing bits (duplication of bits via Expansion Permutation matrix).

MIDTERM 1 FALL 2022 • QUESTION 10

Q: Triple DES algorithm uses a data block size of _____ bits.

- a) 56
- b) 64
- c) 112
- d) 168

A: b) 64

Q: Which block cipher mode is insecure because identical plaintext blocks produce identical ciphertext blocks?

- A) Cipher Block Chaining (CBC)
- B) Counter (CTR)
- C) Electronic Codebook (ECB)
- D) Output Feedback (OFB)

A: C) Electronic Codebook (ECB)

Q: For an N -system symmetric cryptographic network, if a new system comes in, ____ new keys are generated, whereas if a system leaves, ____ keys are removed, and each system remembers ____ keys.

- a. $2N$; $N-1$; N
- b. N ; $N-1$; $N-1$
- c. N ; $2N$; $N-1$
- d. $N/2$; $N+1$; $N+1$

A: b. N ; $N-1$; $N-1$

Part 2: Book Questions

Q: How would you decrypt a message in Counter (CTR) mode? Do you see any similarity between CTR mode and stream ciphers?

A:

- **Decryption in CTR mode:** Decryption is performed by running the block cipher in encryption mode on the counter values (starting from the initial counter/IV) to generate the key stream, and XORing the key stream with the ciphertext blocks.
- **Similarity to Stream Ciphers:** CTR mode generates a continuous key stream independently of the plaintext blocks and XORs it with the input, functioning like a stream cipher.

Q: Distinguish between a block cipher and a stream cipher in terms of operation, speed, and error propagation.

A:

- **Block Cipher:** Processes input in fixed-size blocks (e.g., 64 bits in DES, 128 bits in AES). A single bit error in ciphertext corrupts the entire decrypted block.
- **Stream Cipher:** Encrypts digital data streams continuously one bit or byte at a time. It operates faster and has no error propagation across subsequent bytes.

Q: What are the four core transformations executed in each main round of the Advanced Encryption Standard (AES), and what security function does each perform?

A:

- **SubBytes:** Nonlinear byte substitution via S-Box table lookup; provides confusion.
- **ShiftRows:** Circular byte shifting across rows of the state matrix; provides permutation/diffusion.
- **MixColumns:** Matrix multiplication transforming columns over $GF(2^8)$; provides diffusion across all bytes in a column.
- **AddRoundKey:** Bitwise XOR addition of the 128-bit state matrix with the 128-bit subkey generated from the key schedule.

Q: Explain the operation of Triple DES (3DES) with two keys versus three keys. Why is the sequence Encrypt-Decrypt-Encrypt (EDE) used instead of Encrypt-Encrypt-Encrypt (EEE)?

A:

- **Key Options:** With 2 unique keys (K_1, K_2), the effective key length is 112 bits. With 3 unique keys (K_1, K_2, K_3), the key length is 168 bits.
- **Why EDE:** Using Encrypt-Decrypt-Encrypt (EDE) allows 3DES to be backward compatible with single DES when setting $K_1 = K_2 = K_3$ (since encryption followed by decryption with the same key cancels out).

Q: Why is Electronic Codebook (ECB) mode unsuitable for encrypting structured data or images containing large areas of identical content?

A: In ECB mode, every identical block of plaintext is encrypted into an identical block of ciphertext under the same key. Consequently, underlying structural patterns and shapes in the input remain clearly visible in the output ciphertext.